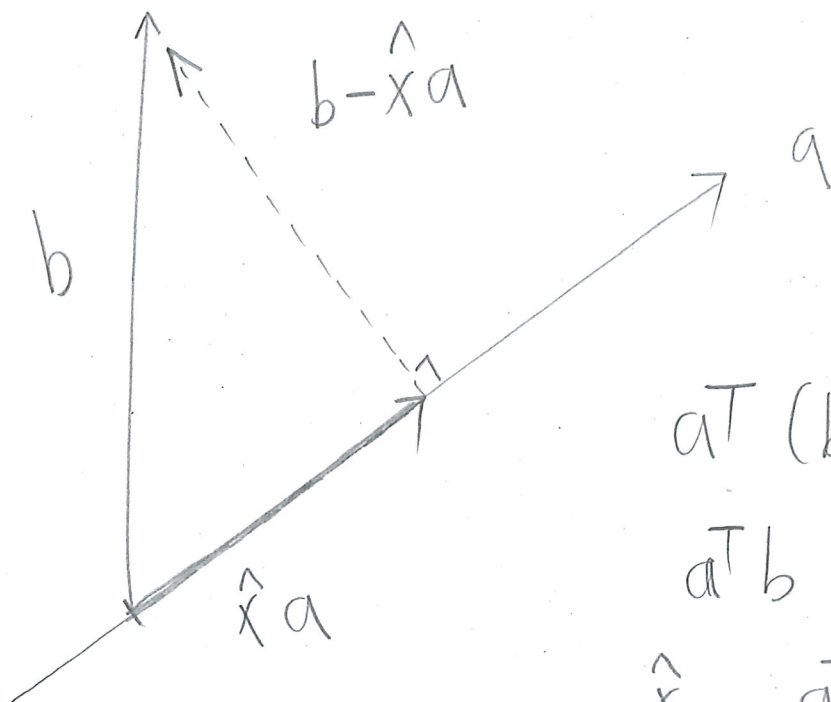




$$a^T (b - \hat{x}a) = 0$$

$$a^T b = \hat{x} a^T a$$

$$\hat{x} = \frac{a^T b}{a^T a}$$

$$p = \hat{x}a = \frac{a^T b}{a^T a} a = \left(\frac{aa^T}{a^T a} \right) b$$

$$= P b$$